

Artificial Intelligence

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Learning Outcomes of Lecture 01

- Introduction to COMP4431
 - Understand the motivation of the lecture teaching
 - Only focus on the discussion of the techniques that have been used widely in industry
 - Only focus on the initial version of the algorithm to emphasize the key idea
 - Understand the motivation of the lab teaching
 - TAs will teach the eight labs and one help session
 - Share the benefits brought by AI
 - New learning and working approach with AI
- Feedback to the Quiz
 - Open book
 - Electronic device are not allowed (except for non programmable calculator)
 - The purpose of the quiz is to help you prepare the final exam
 - Final exam is close book
- History of artificial intelligence

Learning Outcomes of Lecture 01

- What is artificial intelligence
 - A science and a set of computational technologies that are inspired by—but typically operate quite differently from—the ways people use their nervous systems and bodies to sense, learn, reason, and take action
- Birth of artificial intelligence
 - Term coined by McCarthy in 1956 at Dartmouth Conference
 - Search technologies
 - Natural language processing
 - Computer vision
- Golden Time
 - The knowledge revolution
 - The rise of symbolism that symbolic logic is claimed to explain the character of mind
- Winter
 - The difficulty to reproduce general intelligence
- Struggle
 - The rise of machine learning
 - Support vector machines
- Revival
 - Deep learning
 - The success of connectionism
 - The rise of generative AI

Agenda

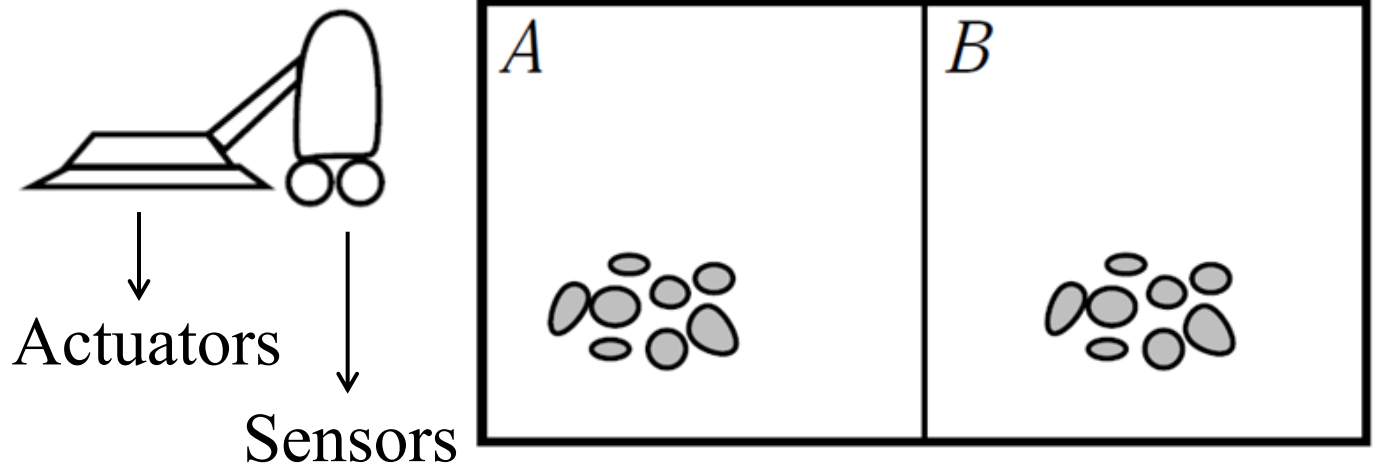
- Review of rational agent
- How to solve the real world problem by rational agent
 - Case study
 - Problem formulation
 - Breadth-first search
 - Depth-first search
- Informed search
 - A* search
 - Heuristics
 - Domination of heuristic functions
- Search strategy

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Agent

- Agents in artificial intelligence
 - Agents denotes artificial entities that are capable of perceiving their surroundings using sensors, making decisions, and then taking actions in response using actuators
- Example
 - Vacuum-Cleaner
- How to perceives the environment
 - which room, clean or dirty
- How to make the decision
 - move right or left, suck the dust
- How to make the action



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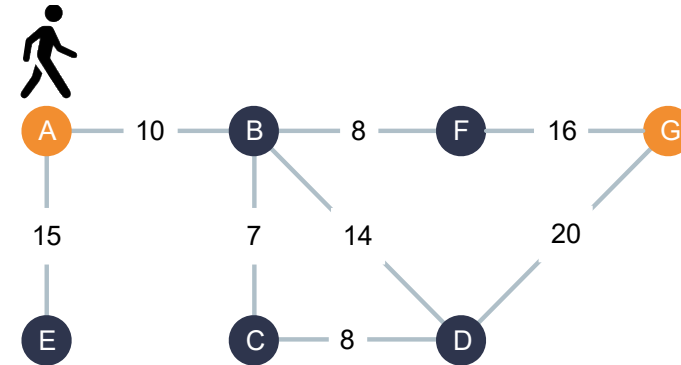
Scenario

- Students visit scientific park
 - ❑ Use intelligent information system to aid arrangement
- Real world problem
 - ❑ Students are in building A
 - ❑ Students want to go to building G
 - ❑ Find a path from building A to building G
 - ❑ Find a shortest path from building A to building G
 - ❑ Find a shortest path from any building to another building



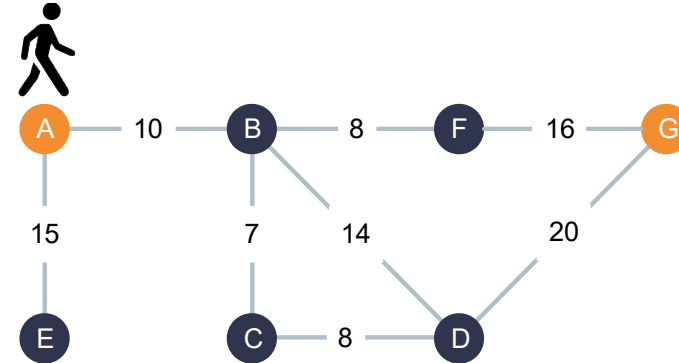
Represent the Map using Graph

- Abstract the map to the graph
 - Ignore the detailed information
- Node
 - Location of the building
- Edge
 - The path connecting two buildings
- Value
 - The time walking from one building to another



Solve the Problem using Rational Agent

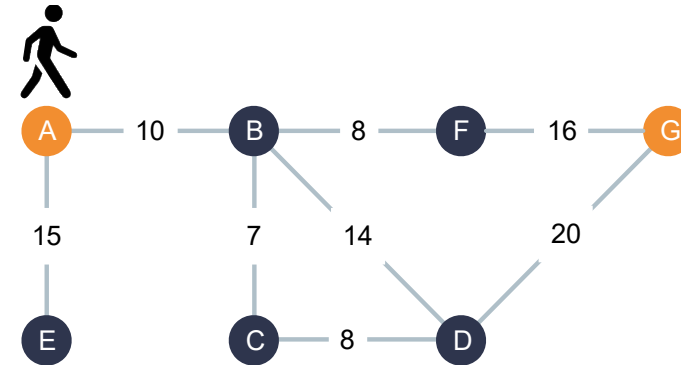
- How to perceive the environment
 - Which kind of information should be collected?
 - Which kind of information could be collected?
 - Which kind of information should be kept?
 - Which kind of information could be kept?
- How to make the action
 - Which kind of action could be made?
 - Which kind of action should be made?
- How to optimize the performance?
 - Could we find the solution?
 - Could we find the optimal solution?
- How to make the decision
 - Check the target
 - Check the cost



- An agent is an entity that perceives and acts
- An agent function is to determine actions from percept histories:
$$[f: \mathcal{P}^* \rightarrow \mathcal{A}]$$
- For any given class of environments and tasks, we seek the agent (or class of agents) with the best performance

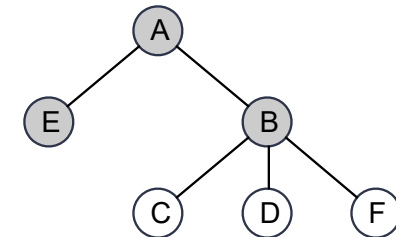
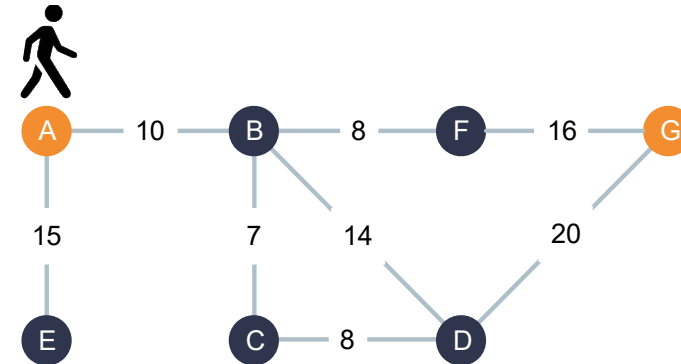
Problem Formulation

- Perception
 - State: $\text{In}(B)$
 - Initial state: $\text{In}(A)$
 - Goal state: $\text{In}(G)$
- Action
 - Decision result $\text{Go}(B)$
 - Solution: $\{\text{Go}(B), \text{Go}(F), \text{Go}(G)\}$
- Transition model
 - Action leads to the state transition
 - Result $(\text{In}(A), \text{Go}(B)) = \text{In}(B)$
- Cost
 - If optimize the performance
 - $c(\text{In}(A), \{\text{Go}(B)\}, \text{In}(B)) = 10$
 - The sum of the costs of the individual actions along the path
 - $c(\text{In}(A), \{\text{Go}(B), \text{Go}(F)\}, \text{In}(F)) = 10 + 8 = 18$



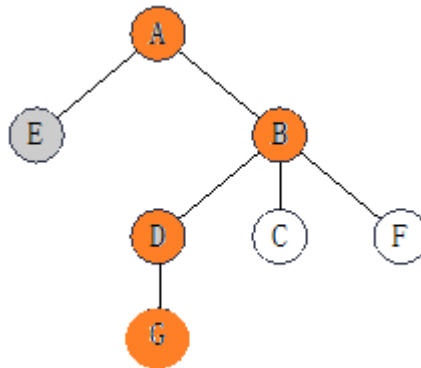
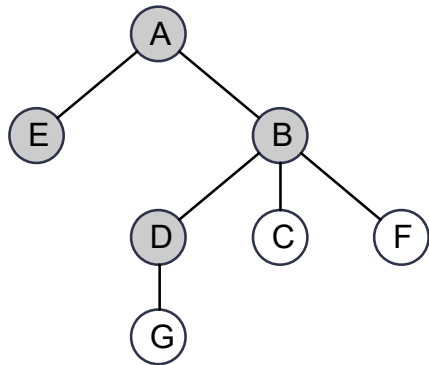
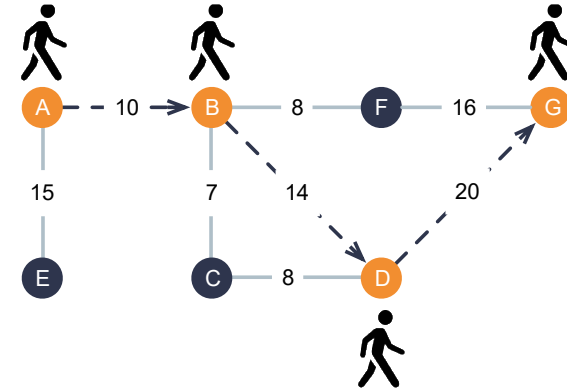
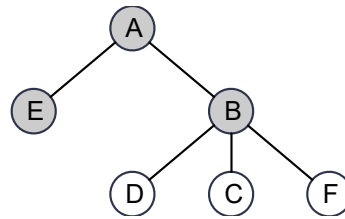
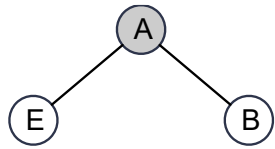
Any Existing Methods to Model the Problem

- Solution
 - An action sequence
- Search algorithm
 - Considering various possible action sequences
- Search tree
 - Nodes: states in the state space
 - Root: initial state
 - Branches: actions
- Expanding the current state
 - Apply each legal action to the current state, thereby generating a new set of states
 - Add branches from the parent node leading to child nodes
- Essence of search
 - Following up one option now and putting the others aside for later
 - In case the first choice does not lead to a solution



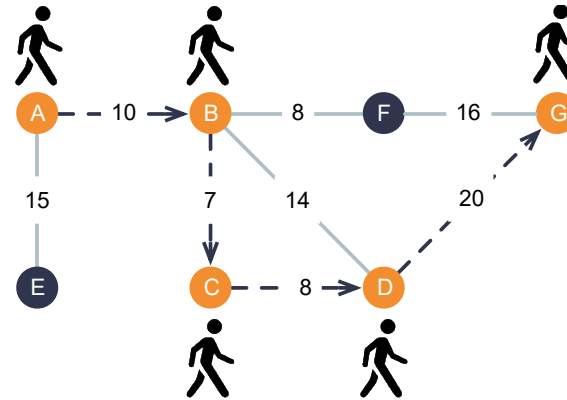
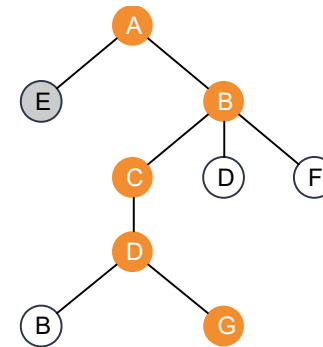
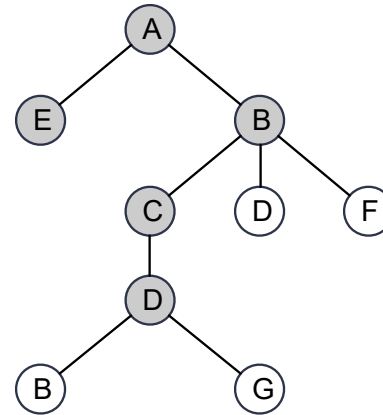
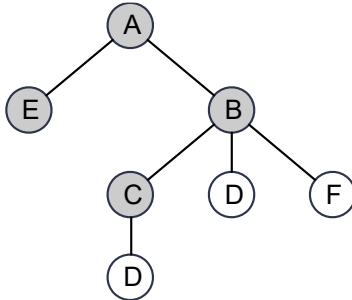
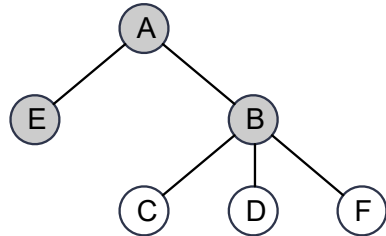
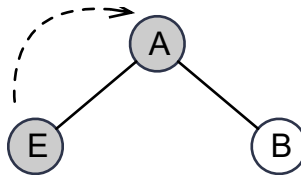
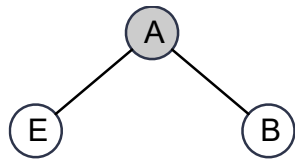
Breadth-first Search

- Expand the shallowest unexpanded node
 - Cost = $10 + 14 + 20 = 44$



Depth-first Search

- Expand the deepest unexpanded node
 - Cost = $10 + 7 + 8 + 20 = 45$

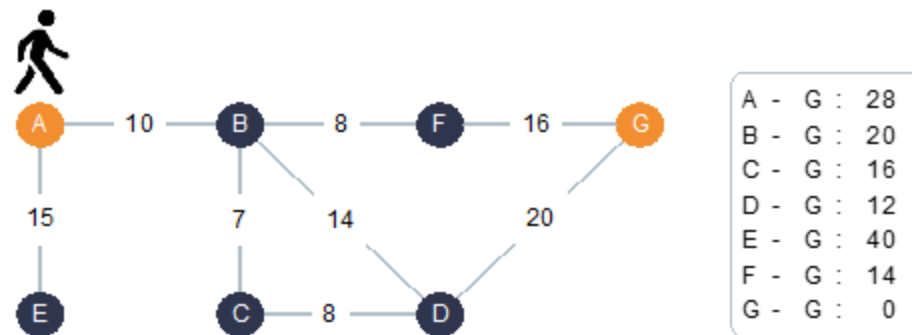


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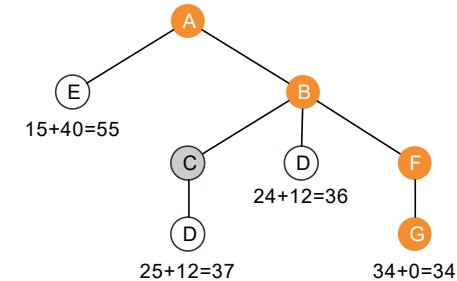
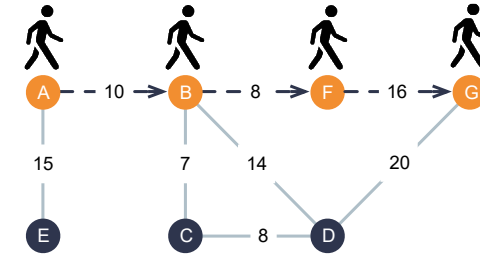
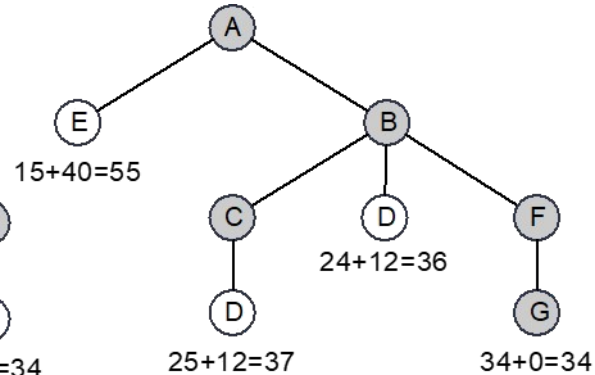
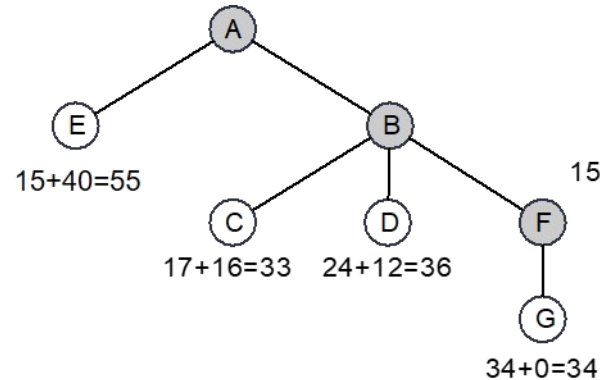
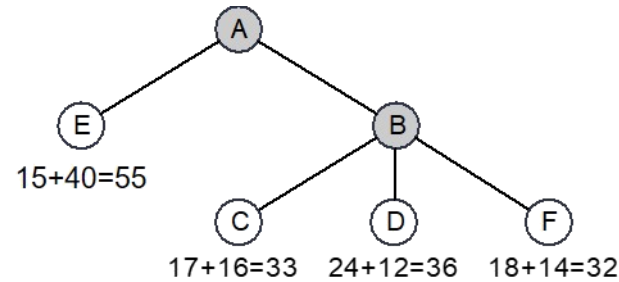
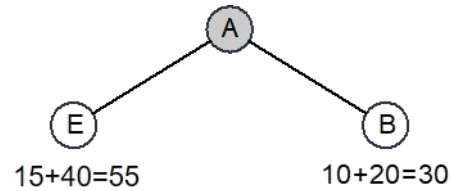
Uninformed Search and Informed Search

- Uninformed search
 - The strategies have no additional information about states beyond that provided in the problem definition
- Informed search
 - The strategies know whether one non-goal node is “more promising” than another
 - Heuristic information represents the relative suitability of a solution among its peers based on the intuition, experience or common sense



A* Search

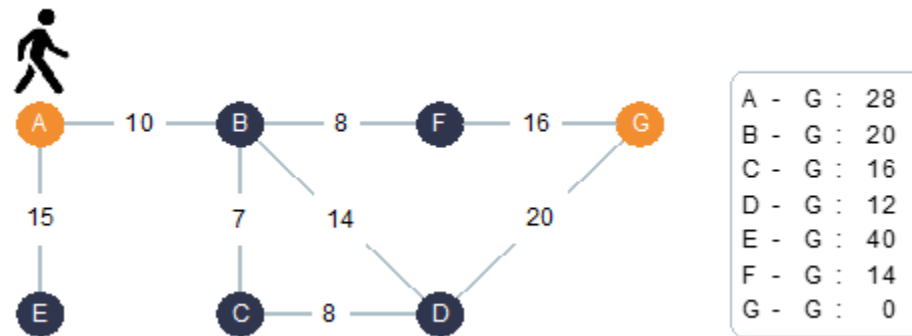
- Evaluation function: $f(n) = g(n) + h(n)$.
 - $g(n)$: the cost so far to reach the node n
 - $h(n)$: the estimated cost to goal from the node n
 - $f(n)$: the estimated total cost of path through the node n to goal



A - G	: 28
B - G	: 20
C - G	: 16
D - G	: 12
E - G	: 40
F - G	: 14
G - G	: 0

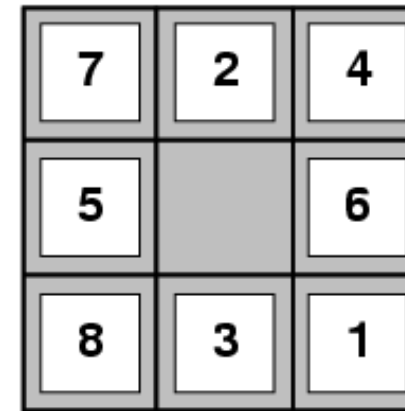
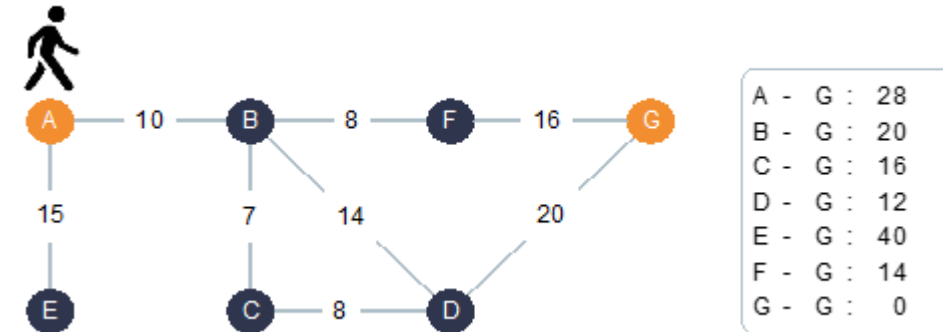
Heuristics

- Heuristics
 - A problem-solving method
 - Uses shortcuts to produce good-enough solutions given a limited time
 - Solution may not necessarily be optimal solution
- Heuristic function in search algorithms
 - A function that ranks alternatives based on available information
- An admissible heuristic never overestimates the cost to reach the goal
 - A heuristic $h(n)$ is admissible if for every node n , $h(n) \leq h^*(n)$, where $h^*(n)$ is the true cost to reach the goal state from n

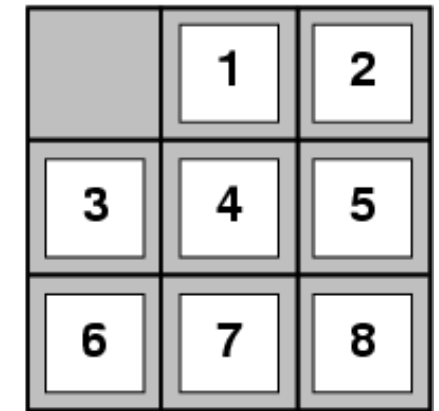


Domination of Heuristic Functions

- If $h_2(n) \geq h_1(n)$ for all n (both admissible)
 - then h_2 dominates h_1 , which indicates that h_2 is better for search
- Domination translates directly into efficiency
 - A* using h_2 will never expand more nodes than A* using h_1
- $h_2(S)$ and $h_1(S)$ for all both admissible
 - $h_1(S) = 8$
 - $h_2(S) = 18$
- A* using h_2 will never expand more nodes than A* using h_1
 - $A^*(h_1) = 39,135$ nodes
 - $A^*(h_2) = 1,641$ nodes



Start State



Goal State

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Search Strategy

- Essence of search
 - ❑ Following up one option now and putting the others aside for later
 - ❑ In case the first choice does not lead to a solution
- Search strategy
 - ❑ Pick the order of node expansion
- Strategies are evaluated along the following dimensions:
 - ❑ completeness: does it always find a solution if one exists?
 - ❑ time complexity: number of nodes generated
 - ❑ space complexity: maximum number of nodes in memory
 - ❑ optimality: does it always find a least-cost solution?

